

## **WP5.3 — Deliverables with Sub-Tasks**

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### **D1 — Data Ingestion & Dataset Preparation**

#### **Objective**

Prepare modeling-ready datasets from WP5.1 or sample setups.

#### **Sub-Tasks**

- Define required process signals and KPIs
- Identify input/output variables for SI
- Define data schema and naming conventions
- Develop data ingestion pipeline/interface
- Synchronize timestamps across signals
- Handle missing/noisy data
- Data filtering and preprocessing
- Feature extraction and normalization
- Dataset labeling (defects/events)
- Generate train/validation/test datasets
- Dataset quality assessment

#### **Output**

- Structured modeling-ready dataset
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### **D2 — System Identification Framework**

#### **Objective**

Develop behavioral/process model from acquired data.

## Sub-Tasks

- Define modeling strategy
  - physics-based
  - ML-based
  - hybrid modeling
- Select SI tools/frameworks
  - MATLAB SI toolbox
  - PyTorch/TensorFlow
- Define model inputs/outputs
- Implement baseline models
- Train and calibrate models
- Hyperparameter tuning
- Compare modeling approaches
- Build reusable SI workflow
- Model version management

## Output

- Identified behavioral model
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# D3 — Offline Model Validation Framework (Validation Loop 1)

## Objective

Validate identified model before deployment/use.

## Sub-Tasks

- Train/test split definition
- Define validation KPIs
- Compute prediction accuracy metrics
- Residual/error analysis
- Sensitivity testing

- Stability checks
- Model confidence estimation
- Validation reporting
- Define model acceptance criteria

## Output

- Validated behavioral model
  - Validation metrics/report
- 

# D4 — Behavioral Digital Twin Framework

## Objective

Use identified model for virtual process simulation.

## Sub-Tasks

- Define DT architecture
- Integrate identified model into DT
- Build process simulation workflow
- Define virtual process inputs
- Develop prediction pipeline
- Implement scenario execution framework
- Simulate process behavior
- Generate predicted outputs
- KPI prediction framework
- Define DT interfaces/APIs

## Output

- Behavioral digital twin environment
  - Predicted process outputs
-

# **D5 — Process Intelligence, Simulation & Optimization Framework**

## **Objective**

Provide framework for simulation, process analysis, and optimization studies.

## **Sub-Tasks**

- Develop what-if simulation capability
- Parameter variation studies
- Sensitivity analysis workflow
- KPI correlation framework
- Root-cause analysis workflow
- Bottleneck analysis framework
- Define optimization objectives
- Define optimization constraints
- Implement optimization workflow
- Parameter recommendation logic
- Multi-objective optimization studies
- Process insight generation
- Reporting/dashboard interfaces

## **Output**

- Simulation & optimization framework
  - KPI insights
  - Optimization recommendations
- 

# **D6 — Virtual Commissioning Integration**

## **Objective**

Validate process and recommendations virtually.

## Sub-Tasks

- VC environment integration
- Process feasibility checks
- Constraint validation
- Virtual process execution
- Collision/reachability verification
- Validate recommended parameter sets
- Generate VC validation reports
- Integration with DT outputs

## Output

- Virtual validation results
- 

# D7 — WP5.1 Visualization Interface

## Objective

Send predictive intelligence outputs to WP5.1.

## Sub-Tasks

- Define interface/API with WP5.1
- Define prediction output schema
- Define KPI visualization data structure
- Export predicted outputs
- Export model health indicators
- Export optimization insights
- Alarm/alert generation support
- Dashboard integration support
- Traceability metadata support

## Output

- Visualization-ready predictive outputs
-

# D8 — Online Validation & Continuous Learning Framework

## Objective

Continuously improve model using runtime data.

## Sub-Tasks

- Predicted vs actual comparison workflow
- Runtime error computation
- Drift detection framework
- Model performance monitoring
- Retraining trigger logic
- Incremental learning workflow
- Dataset update mechanism
- Re-validation workflow
- Continuous improvement pipeline

## Output

- Continuous learning workflow
  - Model improvement indicators
- 

# D9 — MES / Traceability Interface Support

## Objective

Provide integration interfaces for MES and traceability.

## Sub-Tasks

- Define MES integration interfaces
- Define optimization result export format
- Store process metadata

- Store model metadata
- Traceability record management
- Process history linkage
- Reporting interface support

## Output

- MES/traceability integration support
- 

# Very Useful Management Summary

| Deliverable | Main Focus                        |
|-------------|-----------------------------------|
| D1          | Data preparation                  |
| D2          | Model development                 |
| D3          | Model validation                  |
| D4          | Digital twin simulation           |
| D5          | Analysis & optimization framework |
| D6          | Virtual validation                |
| D7          | Monitoring integration            |
| D8          | Continuous learning               |
| D9          | MES & traceability support        |







These are the approximate dates and can. You give the subtasks date aligning with these

# **D1 — Modeling Data Preparation (16-Jun-2026 → 21-Aug-2026)**

## **D1.1 Sample mTorres Data Collection & KPI Identification**

### **Subtasks**

- Identify required process signals and KPIs
- Define data schema and tags
- Collect historical/sample machine data
- Validate signal availability
- Create initial dataset structure

### **Timeline**

16-Jun-2026 → 10-Jul-2026

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## **D1.2 Data Quality & Preprocessing**

### **Subtasks**

- Timestamp synchronization
- Missing data handling
- Noise filtering
- Outlier detection
- Data normalization
- Feature extraction

### **Timeline**

11-Jul-2026 → 05-Aug-2026

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## **D1.3 Modeling Dataset Finalization**

### **Subtasks**

- Generate train/validation/test datasets
- Dataset labeling
- Final data validation

- Curated dataset release

### **Timeline**

06-Aug-2026 → 21-Aug-2026

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## **D2 — Modeling Data Processing from WP5.1 (1-Oct-2026 → 22-Jan-2027)**

### **D2.1 WP5.1 Interface Definition**

#### **Subtasks**

- Define data exchange interface
- Define ingestion APIs/connectors
- Define data formats/schema

### **Timeline**

01-Oct-2026 → 25-Oct-2026

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### **D2.2 Data Ingestion Pipeline Development**

#### **Subtasks**

- Build ingestion workflow
- Implement synchronization logic
- Implement preprocessing pipeline
- Logging/error handling

### **Timeline**

26-Oct-2026 → 10-Dec-2026

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### **D2.3 Dataset Structuring & Validation**

#### **Subtasks**

- Dataset quality checks
- Curated dataset generation
- Validation against WP5.1 signals
- Documentation

### **Timeline**

11-Dec-2026 → 22-Jan-2027

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## **D3 — Defect / Design Model Development (1-Apr-2027 → 18-Jun-2027)**

### **D3.1 KPI / Process Correlation Study**

#### **Subtasks**

- KPI interaction analysis
- Process variable mapping
- Initial process understanding
- Correlation studies

### **Timeline**

01-Apr-2027 → 22-Apr-2027

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### **D3.2 Defect / Process Mapping**

#### **Subtasks**

- Defect-to-process mapping
- Label refinement
- Root-cause interaction study
- Sensitivity identification

### **Timeline**

23-Apr-2027 → 18-May-2027

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## **D3.3 Feature Engineering & Dataset Refinement**

### **Subtasks**

- Feature selection
- Derived feature generation
- Dataset balancing
- Curated training dataset creation

### **Timeline**

19-May-2027 → 18-Jun-2027

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## **D4 — System Identification with Simulated Data (3-Aug-2026 → 4-Dec-2026)**

### **D4.1 SI Problem Definition**

#### **Subtasks**

- Define model inputs/outputs
- Define target process behavior
- Select SI methodology

#### **Timeline**

03-Aug-2026 → 28-Aug-2026

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### **D4.2 SI Framework Setup**

#### **Subtasks**

- MATLAB/Python SI environment setup
- Define reusable workflows
- Data connectors setup

#### **Timeline**

29-Aug-2026 → 25-Sep-2026

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## D4.3 Initial Behavioral Model Development

### Subtasks

- Baseline model creation
- Parameter estimation
- Training/calibration
- Initial prediction validation

### Timeline

26-Sep-2026 → 30-Oct-2026

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## D4.4 Offline Validation Loop 1

### Subtasks

- Train/test validation
- Residual analysis
- Error metrics computation
- Model tuning

### Timeline

31-Oct-2026 → 20-Nov-2026

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## D4.5 Final Behavioral Model Release

### Subtasks

- Documentation
- Final validation
- Export reusable model
- Model readiness review

### Timeline

21-Nov-2026 → 04-Dec-2026

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# **D5 — System Identification Improvement with 5.1 Data (19-Jun-2027 → 3-Sep-2027)**

## **D5.1 Real Data Integration**

### **Subtasks**

- Integrate WP5.1 datasets
- Align real signals with SI workflow
- Dataset synchronization

### **Timeline**

19-Jun-2027 → 10-Jul-2027

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## **D5.2 Model Retraining**

### **Subtasks**

- Retrain behavioral model
- Compare simulated vs real behavior
- Improve robustness

### **Timeline**

11-Jul-2027 → 05-Aug-2027

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## **D5.3 Validation Loop 2 (Online Validation)**

### **Subtasks**

- Predicted vs actual comparison
- Runtime error monitoring
- Drift analysis
- Model improvement workflow

### **Timeline**

06-Aug-2027 → 20-Aug-2027

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## **D5.4 Final Improved Model**

### **Subtasks**

- Final validation
- Performance reporting
- Release updated behavioral model

### **Timeline**

21-Aug-2027 → 03-Sep-2027

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## **D6 — Analysis Framework & Digital Twin (1-Oct-2026 → 19-Mar-2027)**

### **D6.1 Behavioral Digital Twin Integration**

#### **Subtasks**

- Integrate identified model into DT
- Define DT execution pipeline
- Define simulation interfaces

#### **Timeline**

01-Oct-2026 → 10-Nov-2026

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### **D6.2 Scenario Simulation Framework**

#### **Subtasks**

- What-if simulation capability
- Parameter variation studies
- Scenario execution logic

#### **Timeline**



11-Nov-2026 → 20-Dec-2026

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## **D6.3 KPI Correlation & Insight Framework**

### **Subtasks**

- KPI correlation workflow
- Sensitivity analysis
- Root-cause analysis framework

### **Timeline**

21-Dec-2026 → 31-Jan-2027

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## **D6.4 Optimization Framework**

### **Subtasks**

- Define optimization objectives
- Constraint definition
- Recommendation logic
- Multi-objective optimization studies

### **Timeline**

01-Feb-2027 → 28-Feb-2027

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## **D6.5 WP5.1 Visualization Integration**

### **Subtasks**

- Predicted output interface
- Model health visualization
- KPI insight export
- Dashboard integration support

### **Timeline**

01-Mar-2027 → 19-Mar-2027

Actually from 5.3 we don't have to do any of these Timestamp synchronization Missing data handling Noise filtering Outlier detection Data normalization Feature extraction This will be done by 5.1 Once we ingest data , we need to prepare the data for system identification purposes

Then your scope becomes much cleaner and more accurate architecturally.

Those activities:

- timestamp synchronization
- filtering
- noise handling
- normalization
- preprocessing

belong to:

## **WP5.1 — Data Processing / Monitoring Layer**

NOT WP5.3.

WP5.3 should assume:

“cleaned and curated process data is available”

Then WP5.3 focuses on:

## **Modeling-oriented preparation**

for:

- SI
- DT
- optimization

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## **Corrected WP5.3 Scope**

Instead of:

Data preprocessing

Use:

Modeling dataset preparation

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## **Corrected D1**

# **D1 — Modeling Data Preparation**

## **Objective**

Prepare WP5.1 curated data for System Identification workflows.

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## **Correct Subtasks**

### **D1.1 Define SI Modeling Variables**

- Define model inputs/outputs
  - Define KPIs for SI
  - Define target process behavior
  - Define process states/labels
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### **D1.2 Modeling Dataset Structuring**

- Select relevant process signals
  - Windowing/sequence preparation
  - Train/validation/test split
  - Dataset formatting for MATLAB/Python SI frameworks
  - Scenario grouping
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## D1.3 Feature & Label Preparation for SI

- Derived feature preparation for modeling
  - Defect/event labeling for learning
  - Behavioral state classification
  - Dataset balancing
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## D1.4 Dataset Validation for SI Readiness

- Validate completeness for modeling
  - Validate SI compatibility
  - Validate model coverage
  - Curated modeling dataset release
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# MUCH BETTER ARCHITECTURE

| WP5.1                  | WP5.3                  |
|------------------------|------------------------|
| Data acquisition       | System identification  |
| Signal synchronization | SI dataset preparation |
| Filtering              | Behavioral modeling    |
| Noise removal          | Digital twin           |
| Preprocessing          | Simulation             |
| KPI calculation        | Optimization           |

This separation is architecturally correct.

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## Important wording change

Instead of:

Data preprocessing

Use:

Model-oriented dataset preparation

or

Dataset preparation for behavioral modeling

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## Updated realistic subtasks

### What WP5.3 should actually do

#### Inputs from WP5.1

Clean synchronized process dataset

#### WP5.3 tasks

Select → structure → label → prepare  
for SI and DT

NOT:

Acquire → filter → clean

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## Best concise wording for roadmap

### Deliverable

Modeling dataset preparation

### Output

SI-ready behavioral modeling dataset

This is probably the most accurate phrasing for your scope.